

AI DASHCAM PROJECT

PCB & Schematic Design Document

Raspberry Pi 4 Model B | Pi Camera Module v2 (IMX219)

Revision: 1.0 | Date: May 2026 | Classification: Internal

MCU / SBC	CAMERA	PCB LAYERS
Raspberry Pi 4B 4GB LPDDR4	Pi Camera v2 Sony IMX219 8MP	4-Layer FR-4 120 × 85 mm

1. Project Overview

This document defines the complete PCB schematic, component selection, pin mapping, net list, and power architecture for the AI Dashcam system based on the Raspberry Pi 4 Model B. The design targets automotive installation (12V DC input) and integrates video capture, edge AI inference, GPS tracking, CAN bus vehicle telemetry, IMU-based event detection, and proximity sensing.

1.1 System Block Diagram Description

The system is organized into the following functional blocks:

- Processing Core: Raspberry Pi 4 Model B (BCM2711, quad-core ARM Cortex-A72 @ 1.8GHz, 4GB LPDDR4)
- Image Capture: Pi Camera Module v2 (Sony IMX219, 8MP) connected via CSI-2 MIPI interface
- Vehicle Interface: MCP2515 SPI CAN Controller + TJA1050 CAN Transceiver → OBD2 port
- Location & Time: Neo-6M GPS (UART) + DS3231 RTC (I2C)
- Motion Detection: MPU-6050 6-axis IMU (I2C) + AM312 PIR sensor (GPIO)
- Proximity Sensing: 2× HC-SR04 Ultrasonic Sensors (GPIO triggered)
- Storage: W25Q128 SPI Flash (16MB buffer) + microSD Card (primary footage)
- Debug Interface: CH340G USB-UART Bridge for serial console access
- Power System: 12V automotive input → MP2307 Buck (5V) → AMS1117 (3.3V)

2. Microcontroller / SBC Specification

2.1 Raspberry Pi 4 Model B – Core Processor

Processor	Broadcom BCM2711, Quad-core ARM Cortex-A72 @ 1.8GHz
RAM	4GB LPDDR4-3200 SDRAM
GPU	VideoCore VI @ 500MHz – H.265 (4Kp60), H.264 (1080p60)
Camera Interface	1× MIPI CSI-2 (2-lane) — 15-pin FPC, up to 4Kp30
GPIO	40-pin: 28× GPIO, 2× SPI, 2× I2C, 2× UART, 2× 5V, 2× 3.3V, 8× GND
SPI Buses	SPI0 (CE0, CE1, CE2) — for CAN controller, SPI Flash, microSD
I2C Buses	I2C1 (GPIO2/3) — for MPU-6050, DS3231
UART	UART0 (GPIO14/15) for CH340G debug; mini-UART for GPS (GPIO22/23)
Power Input	5.0V ±5% via GPIO Pin 2/4; typical 3.0A, max 3.5A
Operating Temp	0°C to +85°C
Form Factor	85mm × 56mm – Pi standard M2.5 mounting (58×49mm pitch)

2.2 Pi Camera Module v2 – Image Sensor

Image Sensor	Sony IMX219 8-megapixel CMOS
Resolution	3280×2464 (still); 1920×1080p30 / 1280×720p60 / 640×480p90 (video)
Interface	MIPI CSI-2 2-lane, 15-pin FPC ribbon cable
Field of View	62.2° × 48.8° (horizontal × vertical)
Focal Length	3.04mm (fixed focus)
Output Formats	Raw Bayer, YUV/RGB, JPEG via onboard ISP
Power	3.3V via Pi CSI port (no separate power rail required)
Connector	15-pin FPC Connector (DF37B-15DS, 1.0mm pitch)

3. Complete Bill of Materials (BOM)

All components listed below are required for the complete AI Dashcam PCB assembly. Quantities are per unit. Footprints are SMD unless marked THT.

#	Component / Module	Part Number	Qty	Package	Purpose
1	Raspberry Pi 4 Model B (4GB)	RPI4-4GB	1	SBC	Main Processing Unit
2	Pi Camera Module v2	SC0023	1	FPC	Primary Video Capture
3	MCP2515 CAN Controller	MCP2515-I/SO	1	SOIC-18	CAN Bus Interface IC
4	TJA1050 CAN Transceiver	TJA1050T	1	SOIC-8	CAN Bus Driver
5	CH340G USB-UART Bridge	CH340G	1	SOIC-16	Serial Debug / UART
6	AMS1117-3.3 LDO Regulator	AMS1117-3.3	2	SOT-223	3.3V Power Rail
7	MP2307 Buck Converter	MP2307DN	1	SOIC-8	12V to 5V DC-DC Conv.
8	GPS Module (Neo-6M)	NEO-6M-0-001	1	Module	Location Tracking
9	DS3231 RTC Module	DS3231SN	1	SOIC-16	Real-Time Clock
10	MPU-6050 IMU	MPU-6050	1	QFN-24	Accel / Gyro Sensor
11	HC-SR04 Ultrasonic Sensor	HC-SR04	2	Module	Front/Rear Proximity
12	IR Motion Sensor (PIR)	AM312	1	TO-92L	Parking Mode Detect
13	SPI Flash Memory (16MB)	W25Q128JVS1Q	1	SOIC-8	Footage Buffer
14	microSD Card Slot (Push-Push)	DM3D-SF	1	SMD	Main Storage
15	USB-C Power Connector	USB-C SMD	1	SMD	12V Input from Car
16	40-Pin GPIO Header	2x20 2.54mm	1	THT	Raspberry Pi Interface
17	15-Pin FPC Connector	DF37B-15DS	1	SMD	Camera Ribbon Cable
18	4.7kΩ Resistor (I2C Pull-up)	CR0402	4	0402	I2C Pull-ups
19	100nF Decoupling Capacitor	CC0402	12	0402	Bypass Caps
20	10µF Bulk Capacitor	TAJB	6	B-case	Power Filtering
21	LED Red – Power	LTST-C190KRKT	1	0402	PWR Indicator

#	Component / Module	Part Number	Qty	Package	Purpose
2 2	LED Green – Status	LTST-C190KGKT	1	0402	System OK Indicator
2 3	LED Blue – GPS Lock	LTST-C190KBKT	1	0402	GPS Fix Indicator
2 4	33Ω Series Resistors (LEDs)	CR0402	3	0402	LED Current Limit
2 5	N-CH MOSFET (Storage Gate)	2N7002	2	SOT-23	microSD Power Gate
2 6	TVS Diode (12V Rail)	SMBJ12A	1	SMB	Reverse Polarity Prot.



HC-SR04 operates at 5V signal level. Use a resistive voltage divider (1kΩ / 2kΩ) on the ECHO line before connecting to the Pi 3.3V GPIO pin.

4. GPIO Pin Connection Map

Every GPIO pin used on the Raspberry Pi 4 40-pin header with the associated BCM signal name, direction, and the net/device it connects to on the PCB.

GPIO Pin	BCM	Signal Name	Dir	Connected To / Net Name
Pin 1	3V3	POWER_3V3	OUT	AMS1117 OUT → 3.3V Rail
Pin 2	5V	POWER_5V	OUT	MP2307 VOUT → 5V Rail
Pin 3	GPIO 2	SDA1 (I2C)	BIDIR	MPU-6050 SDA, DS3231 SDA [NET: I2C_SDA]
Pin 5	GPIO 3	SCL1 (I2C)	BIDIR	MPU-6050 SCL, DS3231 SCL [NET: I2C_SCL]
Pin 6	GND	GND	PWR	Common Ground Plane [NET: GND]
Pin 8	GPIO 14	UART_TX	OUT	CH340G RXD [NET: UART_TX]
Pin 10	GPIO 15	UART_RX	IN	CH340G TXD [NET: UART_RX]
Pin 11	GPIO 17	SPI0_CE1	OUT	W25Q128 CS# [NET: SPI0_CS1]
Pin 15	GPIO 22	GPS_TX	IN	Neo-6M TX to Pi RX [NET: GPS_UART_RX]
Pin 16	GPIO 23	GPS_RX	OUT	Neo-6M RX from Pi TX [NET: GPS_UART_TX]
Pin 18	GPIO 24	MCP_INT	IN	MCP2515 INT Pin [NET: CAN_INT]
Pin 19	GPIO 10	SPI0_MOSI	OUT	MCP2515 SI, W25Q128 DI [NET: SPI0_MOSI]
Pin 21	GPIO 9	SPI0_MISO	IN	MCP2515 SO, W25Q128 DO [NET: SPI0_MISO]
Pin 23	GPIO 11	SPI0_SCLK	OUT	MCP2515 SCK, W25Q128 CLK [NET: SPI0_CLK]
Pin 24	GPIO 8	SPI0_CE0	OUT	MCP2515 CS# [NET: SPI0_CS0]
Pin 26	GPIO 7	SPI0_CE2	OUT	microSD CS# [NET: SPI0_CS2]
Pin 29	GPIO 5	PIR_OUT	IN	AM312 PIR Output [NET: PIR_SIGNAL]
Pin 31	GPIO 6	ECHO_FRONT	IN	HC-SR04 Front ECHO [NET: ECHO_F]
Pin 33	GPIO 13	TRIG_FRONT	OUT	HC-SR04 Front TRIG [NET: TRIG_F]
Pin 35	GPIO 19	ECHO_REAR	IN	HC-SR04 Rear ECHO [NET: ECHO_R]
Pin 37	GPIO 26	TRIG_REAR	OUT	HC-SR04 Rear TRIG [NET: TRIG_R]
Pin 38	GPIO 20	LED_GPS	OUT	Blue LED via 33Ω [NET: LED_GPS]
Pin 40	GPIO 21	LED_STATUS	OUT	Green LED via 33Ω [NET: LED_STATUS]
CSI	CSI Port	CAM_DATA	BIDIR	Pi Camera v2 via FPC [NET: CSI_LANE0-3]



Pins not listed above (GPIO 0, 1, 4, 16, 25, 27) are reserved for future expansion or left NC. Always decouple unused GPIO with a 100nF capacitor.

5. Complete Net List

Every named electrical net in the schematic. Each net connects the listed nodes and must be correctly routed on all PCB layers. Differential pairs and high-speed SPI traces require controlled impedance.

Net Name	Voltage	Type	Connected Nodes
VIN_12V	12V DC	Power	USB-C Conn → TVS Diode → MP2307 VIN
VCC_5V	5.0V	Power	MP2307 VOUT → Pi Pin2 → AMS1117 VIN → USB Peripherals
VCC_3V3	3.3V	Power	AMS1117 VOUT → Pi Pin1 → MCP2515 VCC → CH340G VCC → All 3V3 ICs
GND	0V	Ground	Common ground plane – all ICs, connectors, decoupling caps
I2C_SDA	3.3V	Digital	Pi GPIO2 <-> MPU-6050 SDA <-> DS3231 SDA [4.7k pull-up to 3V3]
I2C_SCL	3.3V	Digital	Pi GPIO3 → MPU-6050 SCL → DS3231 SCL [4.7k pull-up to 3V3]
SPI0_MOSI	3.3V	Digital	Pi GPIO10 → MCP2515 SI → W25Q128 DI
SPI0_MISO	3.3V	Digital	Pi GPIO9 ← MCP2515 SO ← W25Q128 DO
SPI0_CLK	3.3V	Digital	Pi GPIO11 → MCP2515 SCK → W25Q128 CLK
SPI0_CS0	3.3V	Digital	Pi GPIO8 → MCP2515 CS# (active LOW)
SPI0_CS1	3.3V	Digital	Pi GPIO17 → W25Q128 CS# (active LOW)
SPI0_CS2	3.3V	Digital	Pi GPIO7 → microSD Card CS
UART_TX	3.3V	Digital	Pi GPIO14 → CH340G RXD
UART_RX	3.3V	Digital	Pi GPIO15 ← CH340G TXD
GPS_UART_TX	3.3V	Digital	Pi GPIO23 → Neo-6M RX
GPS_UART_RX	3.3V	Digital	Pi GPIO22 ← Neo-6M TX
CAN_INT	3.3V	Interrupt	MCP2515 INT → Pi GPIO24 (active LOW)
CAN_TX	3.3V	Digital	MCP2515 TXCAN → TJA1050 TXD
CAN_RX	3.3V	Digital	MCP2515 RXCAN ← TJA1050 RXD
CANH	2.5-3.5V	Diff Bus	TJA1050 CANH → OBD2 / Vehicle CAN Bus H
CANL	1.5-2.5V	Diff Bus	TJA1050 CANL → OBD2 / Vehicle CAN Bus L
PIR_SIGNAL	3.3V	Analog	AM312 OUT → Pi GPIO5
TRIG_F	3.3V	Digital	Pi GPIO13 → HC-SR04 Front TRIG
ECHO_F	3.3V*	Digital	HC-SR04 Front ECHO → Voltage Div → Pi GPIO6
TRIG_R	3.3V	Digital	Pi GPIO26 → HC-SR04 Rear TRIG
ECHO_R	3.3V*	Digital	HC-SR04 Rear ECHO → Voltage Div → Pi GPIO19
CSI_LANE0-3	0.9V diff	MIPI CSI	Pi CSI Port <-> Camera Module v2 FPC
LED_STATUS	3.3V	Digital	Pi GPIO21 → 33Ω → Green LED → GND

Net Name	Voltage	Type	Connected Nodes
LED_GPS	3.3V	Digital	Pi GPIO20 → 33Ω → Blue LED → GND
LED_POWER	5V	Power	VCC_5V → 33Ω → Red LED → GND (always on)



All SPI nets must be length-matched within ±0.5mm per group. MIPI CSI differential pairs must be matched within ±0.1mm and routed with s

6. Power Distribution Architecture

The dashcam receives 12V DC from the vehicle via USB-C connector. A TVS diode provides reverse polarity and transient protection at the input.

Rail	Voltage / Current	Regulator	Powered Devices
VIN_12V	12V / 3A max	Car 12V (fused)	Input Rail – MP2307, TVS protection
VCC_5V	5.0V / 2.5A	MP2307 Buck	Raspberry Pi 4, USB peripherals, GPS, HC-SR04
VCC_3V3	3.3V / 800mA	AMS1117-3.3 (A)	MCP2515, TJA1050, CH340G, MPU-6050, DS3231, LEDs
VCC_3V3_B	3.3V / 200mA	AMS1117-3.3 (B)	SPI Flash W25Q128, microSD (gated via MOSFET)

6.1 Power Sequence

- 1. 12V arrives from car ignition → VIN_12V rail energizes
- 2. MP2307 Buck steps 12V → 5.0V (VCC_5V) — output cap charges in ~2ms
- 3. AMS1117-3.3 (A) derives 3.3V (VCC_3V3) from VCC_5V for all digital ICs
- 4. AMS1117-3.3 (B) derives 3.3V (VCC_3V3_B) isolated for SPI Flash and gated microSD
- 5. Raspberry Pi 4 powers on as VCC_5V stabilizes — GPIOs output LOW until boot completes
- 6. Software enables microSD gate MOSFET after OS mounts filesystem

6.2 Decoupling Strategy

Place 100nF (X5R, 0402) capacitors at every IC power pin within 0.5mm. Place 10μF bulk capacitors at the input and output of each regulator. Use 1μF at every SPI device VCC pin in addition to 100nF.

7. I2C Bus Address Map

All I2C devices share the I2C1 bus (SDA: GPIO2, SCL: GPIO3) with 4.7kΩ pull-up resistors to 3.3V.

Device	I2C Address	Notes
MPU-6050 IMU	0x68	AD0 pin = LOW; set AD0 HIGH for 0x69
DS3231 RTC	0x68	Address clash with MPU-6050 – use TCA9548A I2C multiplexer
DS3231 EEPROM	0x57	Internal AT24C32 on RTC module
Neo-6M GPS	N/A	UART only – not on I2C bus



Address conflict: MPU-6050 and DS3231 RTC both default to 0x68. Resolve by adding a TCA9548A I2C multiplexer, or use a software I2C

8. PCB Design Specifications

Board Dimensions	120mm × 85mm (matches Pi 4 footprint with overhang)
Layer Count	4-layer stackup (Signal / GND / PWR / Signal)
PCB Material	FR-4, TG 150, 1.6mm thickness
Copper Weight	1 oz outer / 0.5 oz inner
Min Track Width	0.15mm (signal), 0.5mm (power), 1.0mm (GND fills)
Min Via Drill Size	0.3mm drill / 0.6mm pad
Surface Finish	ENIG (Electroless Nickel Immersion Gold)
Solder Mask	Green LPI, both sides
Silkscreen	White, top and bottom, 0.1mm minimum stroke
Impedance Control	90Ω differential for MIPI CSI; 50Ω single-ended for SPI
Operating Temp Range	-20°C to +70°C (automotive grade)
Connectors	USB-C (power in), 40-pin GPIO header, FPC-15 (camera)
Mounting Holes	4× M3 (3.2mm dia) matching Pi 4 hole pattern

8.1 Layer Stackup

Layer	Name	Type	Usage
L1 (Top)	Signal Top	Copper 1oz	Component side – SMD parts, high-speed SPI, UART
L2	GND Plane	Copper 0.5oz	Solid ground reference plane – do not split
L3	PWR Plane	Copper 0.5oz	Power splits: VCC_5V and VCC_3V3 pours
L4 (Bottom)	Signal Bot	Copper 1oz	Secondary routing – MIPI CSI, slow signals

8.2 Critical Routing Rules

- MIPI CSI pairs (L4): 90Ω differential, less than 50mm trace length, no vias through pairs
- SPI bus (L1): 50Ω single-ended, length-match MOSI/MISO/CLK within ±0.5mm per CS group
- Power traces (VCC_5V): Minimum 1.0mm width, flood pour recommended
- GND vias: Stitch GND plane every 5mm around high-speed ICs
- CAN bus (CANH/CANL): Route as differential pair, 120Ω termination at far end
- Crystal / RTC: Keep DS3231 crystal traces short (<8mm), guard ring to GND
- Decoupling caps: Maximum 0.5mm from IC power pin to cap, then via to GND plane
- LED current resistors: Place within 2mm of LED anode

9. Software & Firmware Configuration

9.1 Raspberry Pi OS — /boot/config.txt Parameters

Parameter	Description
<code>dtoverlay=spi=on</code>	Enable SPI0 bus (MCP2515, W25Q128, microSD)
<code>dtoverlay=i2c_arm=on</code>	Enable I2C1 bus (MPU-6050, DS3231)
<code>dtoverlay=i2c_arm_baudrate=400000</code>	I2C fast mode 400kHz
<code>enable_uart=1</code>	Enable UART0 on GPIO14/15 (CH340G debug)
<code>dtoverlay=miniuart-bt</code>	Move BT to mini-UART, free PL011 for GPS
<code>dtoverlay=mcp2515-can0,oscillator=8000000,interrupt=24</code>	CAN on SPI0, 8MHz XO, INT GPIO24
<code>start_x=1</code>	Enable camera module via CSI
<code>gpu_mem=128</code>	Reserve 128MB GPU memory for camera/video
<code>dtoverlay=disable-bt</code>	Disable onboard BT to free UART for GPS

9.2 Key Python Libraries

- `picamera2` — Camera capture and streaming (replaces deprecated `picamera`)
- `python-can` — CAN bus communication via MCP2515 SocketCAN interface
- `smbus2` — I2C communication for MPU-6050 and DS3231
- `gpsd` + `gps3` — GPS daemon interface for Neo-6M NMEA parsing
- `spidev` — SPI Flash (W25Q128) direct access
- `RPi.GPIO` — GPIO control for ultrasonic sensors, PIR, LEDs
- `tf.lite_runtime` — TensorFlow Lite for on-device AI inference (object detection)

10. Safety & Compliance Notes

- Automotive Protection: Install a 3A automotive blade fuse on the VIN_12V line before the PCB connector
- Reverse Polarity: SMBJ12A TVS diode protects VIN_12V; verify polarity before connecting to vehicle
- Voltage Levels: All Pi GPIO is 3.3V – never connect 5V signals directly (use voltage dividers or level shifters)
- HC-SR04 ECHO: Use a 1kΩ / 2kΩ resistor divider to bring 5V ECHO signal down to 3.3V
- Thermal: MP2307 generates heat under load; ensure adequate copper pour or heatsink pad on bottom
- Camera Handling: Handle FPC ribbon with care; do not fold beyond 3mm minimum bend radius
- ESD: Use ESD-safe tools during assembly; add ESD protection on exposed connectors if needed
- FCC/CE: This design is for personal/prototype use; commercial deployment requires regulatory testing

11. Revision History

Rev	Date	Author	Changes
1.0	May 2026	Engineering Team	Initial release – complete schematic, BOM, net list, pin map, power architecture

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